BRAINMATICS

S.0451.26

BRAINDEVS

Muhammad Firdaus Dwivianto

has successfully completed

Systems Analysis & Design Fundamentals

curriculum course

Date Jan 31 - Mar 6, 2026

BRAINMATICS

Irma Nurmala, S.Kom.

Director of Training

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Instructor

Systems Analysis & Design Fundamentals

Curriculum Course (in 40 Hours)

1. **Introduction**
   - 1.1 Background, Motivation & Concepts
   - 1.2 Software Development Life Cycle
   - 1.3 Software Development Methodology

2. **Systems Planning**
   - 2.1 System Request (Praktik)
   - 2.2 Feasibility Analysis (Praktik)
   - 2.3 Software Effort Estimation (Praktik)
   - 2.4 Software Engineering Roles & Cost

3. **Systems Analysis**
   - 3.1 Requirement Gathering Methods
   - 3.2 Use Case Diagram (Praktik)
   - 3.3 Activity Diagram (Praktik) & BPMN (Praktik)
   - 3.4 Sequence Diagram (Praktik)

4. **Systems Design**
   - 4.1 Object-Oriented Paradigm
   - 4.2 Class Diagram (Praktik)
   - 4.3 UI/UX Design (Praktik)
   - 4.4 Data Model (Praktik)
   - 4.5 Deployment Diagram (Praktik)

5. **Systems Implementation**
   - 5.1 Software Construction (Praktik)
   - 5.2 Software Testing (Praktik)
   - 5.3 Software Documentation
   - 5.4 Software Delivery & Change Management (Praktik)

6. **Digital Transformation Laws & Cases**
   - 6.1 Observations, Laws, & Theories
   - 6.2 50 Laws of Software Engineering
   - 6.3 Digital Transformation Components
   - 6.4 Digital Transformation Case Studies